Survey On PRNU- Image Source Verification & Forgery Detection

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Abstract— **Photo-Response Non-Uniformity (PRNU) is a unique sensor noise pattern that can be used as a fingerprint for camera source verification. Prior research has improved PRNU matching under geometric distortions, but practical issues remain when images are post-processed, compressed, or spliced. Traditional methods may fail in such cases, and AI-based solutions lack transparency. A signal-processing approach is implemented to extract PRNU fingerprints, compare them with known device references, and verify image authenticity. Block-wise correlation analysis is used to identify inconsistencies, enabling detection of splicing and tampering. The method is lightweight, explainable, and suitable for mobile camera images, supporting digital forensics and media authenticity verification without reliance on machine learning.**

Keywords— Photo-Response Non-Uniformity, PRNU Fingerprint, Camera Source Verification, Image Forgery Detection, Block-wise Correlation, Digital Forensics, Image Tampering Detection, Explainable Signal Processing.

I. INTRODUCTION

In an era dominated by digital images, verifying an image's authenticity is a critical challenge. Traditional methods relying on EXIF metadata (e.g., camera model, timestamp) are unreliable as this information can be easily removed or manipulated. This project presents a robust, physics-based solution using Photo-Response Non-Uniformity (PRNU).

PRNU is a unique, unforgeable noise fingerprint inherent to every digital camera sensor. It is a stable, multiplicative noise pattern caused by slight imperfections in the silicon sensor's manufacturing process. Since it is embedded in the very pixels of an image, it cannot be easily removed or altered, making it a reliable forensic marker for image source verification and authenticity.

II. LITERATURE SURVEY

In this survey examines how Photo-Response Non-Uniformity (PRNU), along with computational optimization, statistical analysis, and open-source software, enhances digital image forensics while providing accurate and robust source verification and forgery detection.

The details are revealed by following the process.

A Data Collection

Data collection is the foundational step for any forensic system. In the context of PRNU, this involves gathering a diverse dataset of digital images from various sources, including different camera brands and models, to build a robust database for training and testing. The challenge lies in creating a dataset that accurately reflects real-world scenarios, which often include images that have undergone various post-processing operations. Papers like "ReLoc" highlight the importance of collecting both "plain images" (unprocessed) and "distorted images" that have been subjected to post-processing like lossy compression, blurring, or resizing, as these operations can severely distort forgery traces. The "SNIS" paper also emphasizes the need for datasets that include images manipulated by multiple complex tampering operations and post-processing steps to build robust detection models.

B. Noise Extraction

Noise extraction is the critical step of isolating the faint PRNU noise pattern from the image content. The core principle is to use a denoising filter to suppress the image content and retain the residual noise. The extracted noise residual, denoted as W, can be modeled as $W=I(0)K+\Xi$, where $I(0)$ is the actual optical view, K is the camera fingerprint (PRNU), and Ξ is a combination of other noise components. Several papers discuss advancements in this area:

Traditional Denoising**:** Early methods rely on denoising filters to remove the image content and extract the residual noise. These filters aim to separate the PRNU noise from other image components, including quantization noise and scene details.

Deep Learning-based Methods**:** The paper on "Unsupervised Learning-Based Framework for Deepfake Video Detection" discusses the use of a trained deep neural network, referred to as a "noise print," as a CNN-based denoising filter. This

approach learns to generate a noise residual that discards scene content and other non-discriminative information while enhancing the discriminative camera model artifacts. Similarly, the "Face Forgery Detection" paper utilizes a "Learnable SRM" (Steganalysis Rich Model) filter to extract noise features that complement spatial information.

Addressing Post-processing: A major challenge is that postprocessing can introduce "software noise" that pollutes the PRNU fingerprint and interferes with matching. The paper on "ACL" defines this software noise as a persistent artifact introduced by image post-processing that cannot be easily eliminated by conventional denoising. It models a polluted fingerprint as a combination of the hardware fingerprint and software noise, making noise extraction a more complex task.

Handling Distortions: The paper "An Adaptive Method for Camera Attribution Under Complex Radial Distortion Corrections" addresses how geometric transformations like radial distortion corrections can misalign the image grid, making standard PRNU extraction and correlation ineffective. The proposed solution involves processing the image in concentric annuli to locally handle complex distortions and accurately extract the noise.

C. Fingerprint Creation

The goal of this module is to create a robust and reliable "master fingerprint" or "camera reference pattern" by aggregating the noise extracted from multiple images captured by a single device. Averaging multiple noise residuals helps suppress random noise components and enhances the unique, systematic PRNU pattern.

Maximum Likelihood Estimation (MLE): The most common method for fingerprint creation is Maximum Likelihood Estimation (MLE), which estimates the camera fingerprint from a number of noise residuals. This process improves the precision of the fingerprint as the number of images used increases.

Noise print Aggregation**:** The "Unsupervised Learning-Based Framework" paper describes creating a reference noise print by averaging several noise prints extracted from images taken by the same camera model. This approach creates a more robust fingerprint that is effective even when dealing with videos and different camera models.

Handling Videos: For videos, the process of fingerprint creation is adapted by extracting the noise from each frame. The "Unsupervised Learning-Based Framework" suggests a process of segmenting all full-size frames into groups, calculating the noise residuals for each group, and then averaging the residuals to obtain a single PRNU feature for the entire video.

Fingerprint creation involves aggregating noise residuals from multiple images to form a robust PRNU pattern and noise print averaging improves reliability, and for videos, frame-level residuals are averaged to create a unified camera fingerprint.

D. Verification of Samples (Image Verification & Forgery Detection)

This module is the core of the project, where the extracted noise from an unknown image is compared to the created fingerprints to verify its source or detect forgeries.

Image Source Verification

Source verification involves comparing a query image's noise to a known device's reference fingerprint. This comparison is typically done using a test statistic that measures the correlation between the two.

Conventional Test Statistics**:** Standard methods use cross-correlation (CC), normalized cross-correlation (NCC), or the Peak-to-Correlation Energy (PCE) ratio. A high value indicates a match, while a low value suggests the image is from a different device.

Handling Software Noise: The "ACL" paper highlights the failure of conventional statistics when "software noise" is present, as it can cause a high correlation even between fingerprints from different devices of the same model. To address this, it introduces a new test statistic called Fingerprint to Software Noise Ratio (FITS**)**. FITS determines if a correlation peak is caused by the hardware fingerprint or just the software noise, allowing for a robust matching threshold to be set.

Handling Geometric Distortions: The paper on "An Adaptive Method for Camera Attribution" addresses how radial distortion corrections, especially complex ones from professional software like Adobe Lightroom, can make standard PRNU matching fail. It proposes an adaptive algorithm that divides the image into concentric annuli, allowing for a localized and efficient search for the optimal distortion correction parameter. It introduces the cumulative peak of correlation energy (CPCE) to make this process computationally efficient.

Unsupervised Verification: The "Unsupervised Learning Based Framework for Deepfake Video Detection" proposes a fully unsupervised method for video source verification. It uses PRNU of the full-size frame in a first clustering stage to group videos from the same source, regardless of whether they are real or fake.

Forgery Detection

Forgery detection focuses on identifying inconsistencies within a single image or video.

Noise Inconsistency**:** Forgery techniques often involve splicing, copy-move, or removal, which disrupt the natural PRNU pattern of the original image. By locating regions with inconsistent noise characteristics, forgeries can be detected.

Deep Learning-based Methods**:** The paper "SNIS: A Signal Noise Separation-Based Network" formulates forgery detection as a signal noise separation problem. The network

separates the forged region (the "main signal") from the background and post-processing noise (the "noise signal"), making the forgery traces easier to detect. This approach is particularly effective against post-processing operations that typically obscure forgery traces.

Domain Adaptation for Robustness: The "Towards JPEGResistant Image Forgery Detection" paper addresses the challenge of JPEG compression, which can severely degrade forgery detection performance. It proposes a self-supervised domain adaptation network that uses a "Compression Approximation Network" (ComNet) to generate "JPEG-agent images" that mimic general JPEG compression characteristics. The main network then learns to detect forgeries on these simulated images, making it robust to various and unknown JPEG compression quality factors.

Hybrid Feature Fusion: The paper "Face Forgery Detection via Multi-Feature Fusion and Local Enhancement" proposes a network that combines both spatial-domain information and noise-domain information. This dual-stream approach leverages the complementary nature of these features to create a more robust forgery detection system that is less prone to overfitting on specific color textures.

Restoration-Assisted Framework:"ReLoc: A Restoration Assisted Framework" introduces a novel approach where a restoration module is used to recover a high-quality counterpart of a distorted tampered image before it is fed to the forgery detection module. The restored image re-enhances the subtle tampering traces that may have been weakened by post-processing, significantly improving the localization performance.

E Dashboard & Console / Future Directions

The Dashboard and Console component serves as a user-friendly interface for monitoring and interacting with the PRNU-based image forensics system. It provides a high-level overview of system activity through key metrics, such as the number of verified images and detected forgeries [7]. This dashboard also allows users to manage the system, including uploading new images for analysis and managing the database of camera fingerprints. The console's primary function is to present the detailed results of the forensic analysis, including performance metrics like F1-score and AUC, as well as visual aids like heatmaps that pinpoint tampered regions in an image [7].

F. Future Directions

The field of PRNU-based image forensics is continually evolving to address sophisticated digital manipulation techniques. Future research should focus on three key areas. Firstly, improving the robustness of detection algorithms against increasingly complex post-processing attacks like blurring, noise corruption, and geometric distortions. Secondly, advancing unsupervised and hybrid learning methods to overcome the limitations of large, labeled datasets and enhance generalization to new types of forgeries. This includes fusing traditional forensic features like PRNU with modern deep learning models. Finally, exploring emerging forensic artifacts, such as moiré patterns and optical-property-based cues, offers a new avenue for developing multi-layered defense systems against digital media forgery.

III. EXISTING WORK

The existing systems in digital forensics currently utilize a variety of measures to protect image authenticity, but these systems often lack comprehensive integration and real-time responsiveness. Most organizations deploy traditional security tools such as metadata analysis and checksum verification. While these tools provide a baseline level of protection, they often operate in silos, making it difficult to gain a holistic view of an image's integrity. Many organizations rely on digital forensics labs for centralized monitoring, but these labs typically focus on basic threat detection without the advanced capabilities offered by newer technologies like PRNU (Photo-Response Non-Uniformity) analysis and forgery detection solutions. Existing incident response processes may be reactive rather than proactive, resulting in delayed responses to security incidents. Furthermore, compliance with regulatory standards is often managed through periodic audits rather than continuous monitoring, which can lead to vulnerabilities remaining unaddressed. This fragmented approach hampers the ability to effectively detect and respond to emerging threats, leaving organizations at risk of image forgery and compromised authenticity.

V. CONCLUSION

The field of PRNU-based image forensics has demonstrated significant advancements in combating digital image manipulation. By leveraging the unique, inherent noise patterns of digital camera sensors, researchers have developed robust methods for both image source verification and forgery detection. Modern approaches, informed by deep learning and sophisticated signal processing techniques, are proving increasingly effective against complex challenges like post processing noise, geometric distortions, and new forms of generative AI-based forgery. While challenges remain, particularly in the face of evolving manipulation tactics, the foundational principles of PRNU, combined with ongoing innovation, offer a powerful and promising path forward for ensuring the authenticity and integrity of digital media. the field of PRNU-based image forensics has demonstrated significant advancements in combating digital image manipulation. By leveraging the unique, inherent noise patterns of digital camera sensors, researchers have developed robust methods for both image source verification and forgery detection. Modern approaches, informed by deep learning and sophisticated signal processing techniques, are proving increasingly effective against complex challenges like post processing noise, geometric distortions, and new forms of generative AI-based forgery. While challenges remain, particularly in the face of evolving manipulation tactics, the foundational principles of PRNU, combined with ongoing innovation, offer a powerful and promising path forward for ensuring the authenticity and integrity of digital media.

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